OPT example

1	2	3	4	1	2	5	1	2	3	4	5
m	m	m	m	h	h						

4	3	2	1

OPT example

1	2	3	4	1	2	5	1	2	3	4	5
m	m	m	m	h	h	m					

Evict 4 (furthest away in the future)

5	3	2	1

OPT example

1	2	3	4	1	2	5	1	2	3	4	5
m	m	m	m	h	h	m	h	h	h		

3	2	1	5

OPT example

1	2	3	4	1	2	5	1	2	3	4	5
m	m	m	m	h	h	m	h	h	h	m	

4	2	1	5

OPT example

1	2	3	4	1	2	5	1	2	3	4	5
m	m	m	m	h	h	m	h	h	h	m	h

4	2	1	5

Total misses: 6 (this is the optimal!)

Eviction policies in the real world

- All of the eviction policies we surveyed are used by real systems
  - Except for OPT!
- LRU and its approximations are the most common
- Complications:
  - We assumed all items are the same size
  - Things get more complicated when items can have different sizes
    - E.g., Bigger items might cause the eviction of many small items

Scaling Memcache at Facebook

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Facebook Inc.

Abstract: `Memcached` is a well known, simple, in-memory caching solution. This paper describes how Facebook leverages `memcached` as a building block to construct and scale a distributed key-value store that supports the world's largest social network. Our system handles billions of requests per second and holds trillions of items to deliver a rich experience for over a billion users around the world.

1 Introduction

Popular and engaging social networking sites present significant infrastructure challenges. Hundreds of millions of people use these networks every day and impose computational, network, and I/O demands that traditional web architectures struggle to satisfy. A social network's infrastructure needs to (1) allow near real-time communication, (2) aggregate content on-the-fly from multiple sources, (3) be able to access and update very popular shared content, and (4) scale to process millions of user requests per second.

We describe how we improved the open source ver-

however, web pages routinely fetch thousands of key-value pairs from `memcached` servers.

One of our goals is to present the important themes that emerge at different scales of our deployment. While qualities like performance, efficiency, fault-tolerance, and consistency are important at all scales, our experience indicates that at specific sizes some qualities require more effort to achieve than others. For example, maintaining data consistency can be easier at small scales if replication is minimal compared to larger ones where replication is often necessary. Additionally, the importance of finding an optimal communication schedule increases as the number of servers increase and networking becomes the bottleneck.

This paper includes four main contributions: (1) We describe the evolution of Facebook's `memcached`-based architecture. (2) We identify enhancements to `memcached` that improve performance and increase memory efficiency. (3) We highlight mechanisms that improve our ability to operate our system at scale. (4) We characterize the production workloads imposed on our system.

https://www.usenix.org/system/files/conference/nsdi13/nsdi13-final170_update.pdf

TCP/IP
5-Layer Model

How Data Actually Travels Across the Internet

1 Physical	2 Data Link	3 Network	4 Transport	5 Application

Graduate Course in Computer Networking

The 5-Layer Stack

5 — Application

HTTP · HTTPS · SMTP · FTP · SSH · DNS

4 — Transport

TCP · UDP · QUIC · Port numbers

3 — Network

IPv4 · IPv6 · BGP · Routers

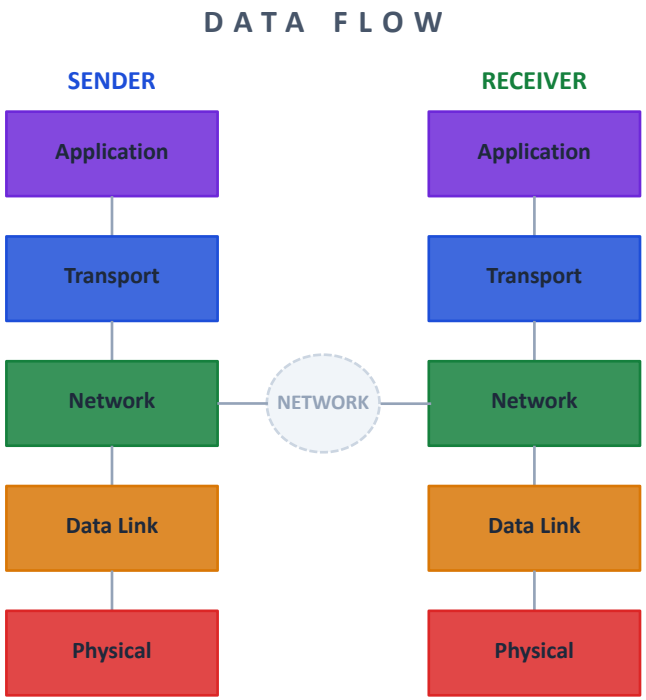
2 — Data Link

Ethernet frames · MAC · ARP · DHCP

1 — Physical

Fiber · Copper · Coax · WiFi · 5G · Satellite · Microwave

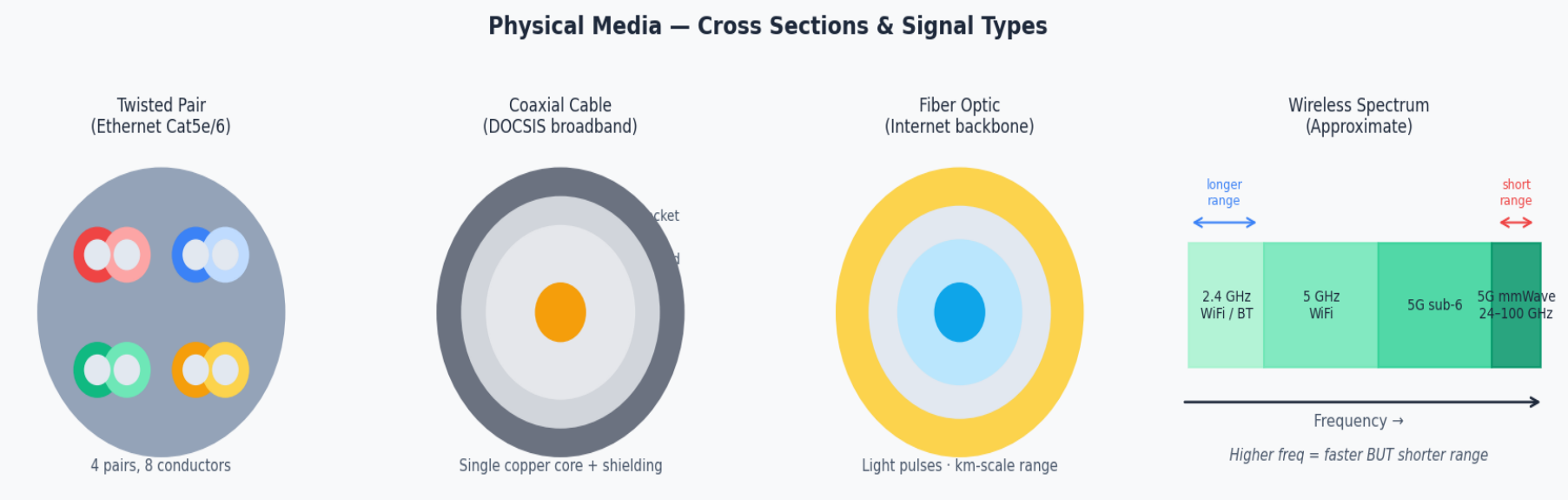
← Encapsulation (sending) · Decapsulation (receiving) →



Layer 1: Physical

L1 Physical

Converts bits to physical signals. The medium determines speed, range, and latency.



Ethernet spans L1 (signaling) + L2 (framing) — the same cable serves both layers

Microwave travels faster than fiber (air vs. glass) — HFT firms use this for trading

GEO satellite: ~600ms latency (physics). Starlink LEO: 20–40ms — viable for apps

KEY CONCEPT Layer 1 is protocol-agnostic — it just moves signals. A broken cable or weak signal kills everything above it regardless of how good your software is.

Layer 2: Data Link

L2 Data Link

Organizes bits into frames · handles local network delivery · MAC addresses identify hardware.

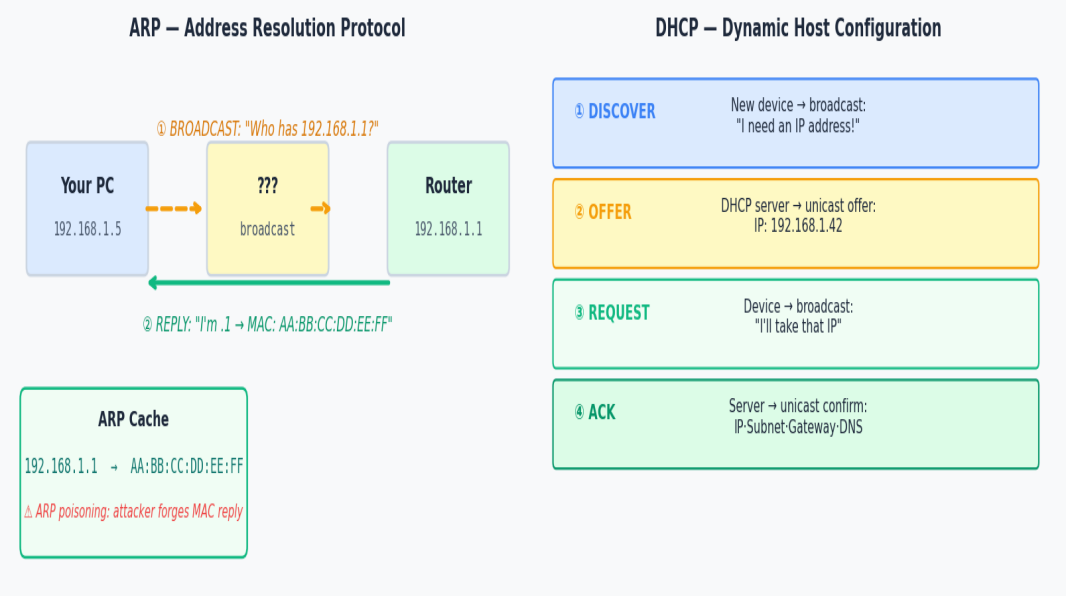
ETHERNET FRAME STRUCTURE

Preamble	Dst MAC	Src MAC	Type	Payload	FCS
8B	6B	6B	2B	46-1500B	4B

MAC Address

3C:22:FB:4A:9E:01

48-bit hardware identifier · burned in at manufacture
Local only — invisible beyond the router
Switches use MAC tables; routers do not



KEY CONCEPT MAC addresses solve local delivery. ARP bridges L2 (MAC) and L3 (IP). DHCP bootstraps new devices automatically. Both are critical infrastructure — and both have well-known attacks.

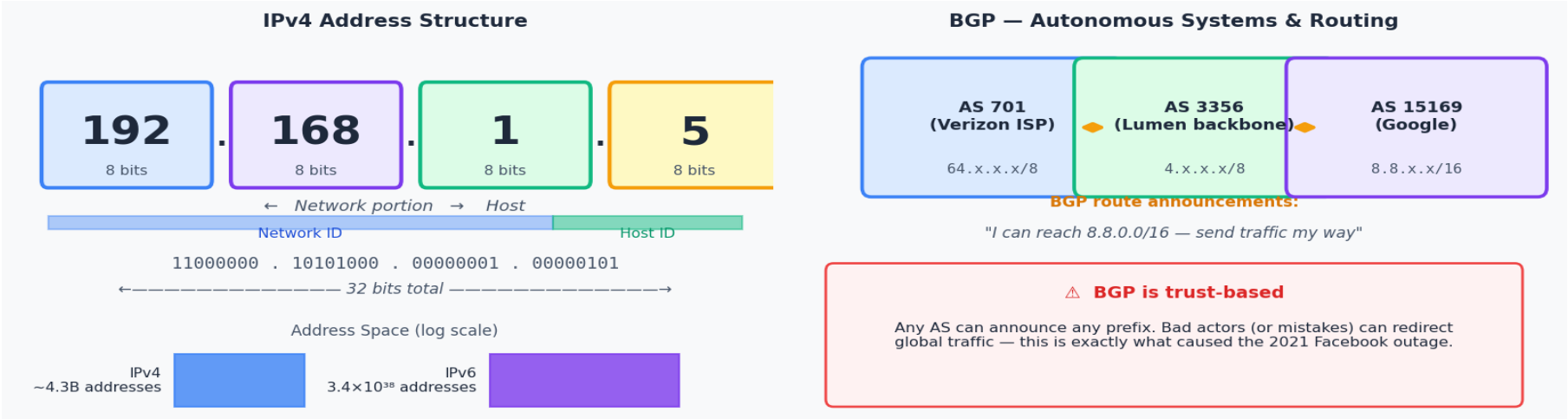
Layer 3: Network

L3 Network

Routes packets across multiple networks using IP addresses. Makes the global internet possible.

IPv4 → 32-bit · ~4.3B addresses · Exhausted (NAT extends it)
IPv6 → 128-bit · 340 undecillion · Eliminates NAT · Dominant on mobile

```
Run traceroute google.com in a terminal to see every router hop in real time.
```



KEY CONCEPT IP addresses are logical and hierarchical — they encode location. No router has a complete map: each makes a local best-hop decision. BGP's trust model is its greatest vulnerability.